

Every Threshold Has a Place: Conserved Spatial Evidence Ledgers for Auditable Ordinal Vision

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Abstract

001 Disease severity is ordinal and is often supported by
002 small, spatially distributed findings, yet most grad-
003 ing networks first collapse an image to a global vec-
004 tor and explain the resulting decision afterward. We
005 introduce Origin, an architecture whose only image-
006 dependent prediction input is a conserved, boundary-
007 indexed spatial evidence ledger. A multiscale con-
008 volutional encoder emits bounded nonnegative atoms
009 for adjacent severity transitions; prerequisite compila-
010 tion makes higher-severity atoms support their lower
011 thresholds. Summing the ledger produces four transi-
012 tion parameters, decoded in our primary implementa-
013 tion by a pure-birth generator into a normalized five-
014 grade posterior. Removing any ledger group and re-
015 playing the same deterministic decoder gives an ex-
016 act frozen-ledger internal intervention, without gra-
017 dients, pixel replacement, re-encoding, or attention
018 renormalization. This guarantee belongs to the con-
019 served ledger, not uniquely to the matrix exponen-
020 tial: an explicitly defined sequential-hazard decoder
021 shares it, whereas the CTMC alone supplies a ho-
022 mogeneous shared-clock semigroup across exposure
023 horizons. On locked, patient-grouped 10-fold Eye-
024 PACS outer cross-validation ($N = 35,126$ pooled out-
025 of-fold predictions), Origin obtains 85.99% accuracy,
026 0.8153 quadratic weighted kappa (QWK), 0.1963 dis-
027 crete MAP-grade mean absolute error (MAE), and
028 0.2496 posterior-expected-grade MAE; fold accuracy
029 is $85.99 \pm 0.44\%$. On complete 5-fold APTOS outer
030 cross-validation, it obtains 84.87% accuracy, 0.9118
031 QWK, and 0.1936 MAP-grade MAE. In a protocol
032 frozen before its ten trainings but after EyePACS
033 fold 9 had already been observed, a retrospective study
034 finds that a pooled continuation-ratio control obtains
035 86.40% accuracy versus 85.66% for Origin and reduces
036 MAP-grade MAE by 0.0145 (paired patient-cluster
037 95% CI [0.0017, 0.0274]), but worsens ECE by 0.0273
038 ([0.0198, 0.0357]). Fine-scale evidence alone loses 1.97

accuracy points ([1.08, 2.88]). Thus the present evi-
dence supports accurate grading and exact ledger re-
play, but not an accuracy advantage of the factoriza-
tion, pixel causality, or lesion semantics.

1. Introduction

Diabetic retinopathy (DR) is a common complication
of diabetes and a major cause of preventable vision loss;
an estimated 103 million adults had DR in 2020, with
the burden projected to rise substantially by 2045 [28].
Clinical screening assigns an ordered severity grade
from no apparent DR through proliferative DR [30].
This ordering is not cosmetic: confusing neighboring
stages and skipping several stages represent different
errors, and severe grades are supported by increasing
or qualitatively stronger retinal findings.

Modern ordinal methods encode order through cu-
mulative thresholds [5, 25], soft targets [7], structured
embeddings [17, 18], contrastive margins [21, 23], or la-
bel sequences [29, 31]. These methods improve the out-
put geometry, but their decision is normally made from
a pooled representation. Conversely, post-hoc maps
such as Grad-CAM [24] may localize a prediction, but
are not the prediction mechanism; moreover, param-
eter and data randomization tests reveal that some pop-
ular saliency variants can be insensitive to the learned
model [1]. Even intrinsically spatial models usually ag-
gregate local scores into class logits [8, 13] or attention
tokens [26]. What remains underexplored is an ordi-
nal object that addresses evidence at every adjacent
threshold and can be intervened on without changing
unselected computation. We seek three properties to-
gether:

1. a valid joint posterior over ordered grades;
2. a prediction path composed exclusively of spatial
evidence; and
3. an intervention whose reported effect is exactly the
effect inside that same prediction circuit.

We propose Origin to couple these properties through a conserved ordinal evidence ledger. The model converts multiscale feature cells into bounded, nonnegative contributions to the four adjacent transitions of a five-grade scale. These entries are the complete image-dependent input to a declared ordinal decoder. Our primary decoder is a pure-birth CTMC, but the ledger contract is decoder-agnostic: subtraction followed by the same deterministic decoder is exact for either the CTMC or a sequential-hazard control. The explanation is therefore not a visualization of a hidden classifier; it is an addressable decomposition of the quantity from which the posterior is computed.

Our contributions are:

- Boundary-indexed conserved evidence. Origin makes a nonnegative receptive-field-indexed ledger the exclusive image-dependent input to every ordinal threshold; no pooled classifier or prediction bypass exists. A null-simplex head, prerequisite compiler, and conservation law make evidence cumulative, bounded, and resolution independent.
- Exact grouped ledger interventions. Deleting or retaining declared ledger groups and replaying the identical decoder gives the literal posterior change under that frozen-ledger intervention; no backward pass, pixel replacement, or renormalization is involved.
- Explicit decoder contracts. Both the CTMC and matched sequential-hazard decoder yield normalized fixed-horizon posteriors whose cumulative tails are coordinatewise monotone in the rates, as well as exact ledger replay. The CTMC-specific property is a homogeneous shared-clock Markov semigroup across horizons, whose utility is not tested by one endpoint grade per image.
- Locked ordinal evaluation. We report complete 5-fold APTOS and patient-grouped 10-fold EyePACS outer cross-validation, including QWK, MAE, calibration, balanced accuracy, and minority-grade recall rather than accuracy alone. A separate ten-arm, patient-paired fold-9 mechanism study reports both favorable and falsifying evidence for the architectural components without blending it into the full-CV estimate.

2. Related Work

Ordinal image prediction. Deep ordinal learning has been formulated as unimodal output distributions [3], soft labels [7], rank-consistent cumulative or conditional classifiers [5, 14, 25], probabilistic embeddings [18], and consistent representations [17]. Ord2Seq generates a label sequence [29]; CLOC learns boundary-

dependent contrastive margins [21]; and recent DR systems combine ordinal learning with contrastive objectives [23] or autoregressive diffusion [31]. Directed ordinal diffusion regularization shapes a feature space according to a forward disease trajectory [6]. In contrast, our pure-birth construction is an image-level algebraic decoder: its unit exposure is not elapsed biological time, it does not model a patient's longitudinal progression, and it produces the joint posterior in one matrix exponential.

Spatial and intrinsic interpretation. Gradient maps [24] explain a trained predictor after the fact. Lesion-aware transformers jointly grade DR and discover regions from image labels, but their localization remains a learned auxiliary mechanism [27]. Concept bottlenecks expose semantic intermediate variables [15], but normally require concept supervision and do not by themselves provide local evidence. Additive MIL makes regional contributions to a pooled prediction explicit [13]. Sparse disease grading adapts BagNet to produce fine, faithful class-evidence fields for small DR lesions [8], while ordinal MIL uses selective aggregation tokens for threshold-specific evidence [26]. These are closest in spirit, but Origin differs in the object being decomposed: each local entry is a nonnegative vector of adjacent-state generator rates, and the same stored entries can be removed and replayed through the exact ordinal decoder.

Generator view. For a continuous-time Markov chain, a valid generator defines a stochastic transition matrix through the matrix exponential [16]. Li and Lin superpose transition rates derived from specialized classifiers and use the CTMC equilibrium distribution to combine their predictions [19]. DiDiCM instead learns a discrete diffusion process over general class labels [4]. Origin differs from both: its primary forward-only pure-birth chain maps a conserved local evidence ledger to a finite-time ordinal posterior. We use the generator as a constrained probability layer rather than as a temporal disease model. Importantly, exact ledger replay is not CTMC-specific; we compare a sequential-hazard map applied to the identical ledger. The distinguishing CTMC contract is instead the homogeneous semigroup $\exp((t+r)Q) = \exp(tQ)\exp(rQ)$ across exposure horizons [16]. Scaling and squaring is a standard route to stable matrix exponentials [12]; our small structured generator permits a fixed, audited FP64 implementation.

ORIGIN: Ledger-Exclusive Ordinal Severity Grading

Spatial evidence is the computation: bounded local transition rates $\rightarrow$ one conserved ledger $\rightarrow$ one declared ordinal decoder $\rightarrow$ one grade posterior.

NO POOLED CLASSIFIER / NO BYPASS

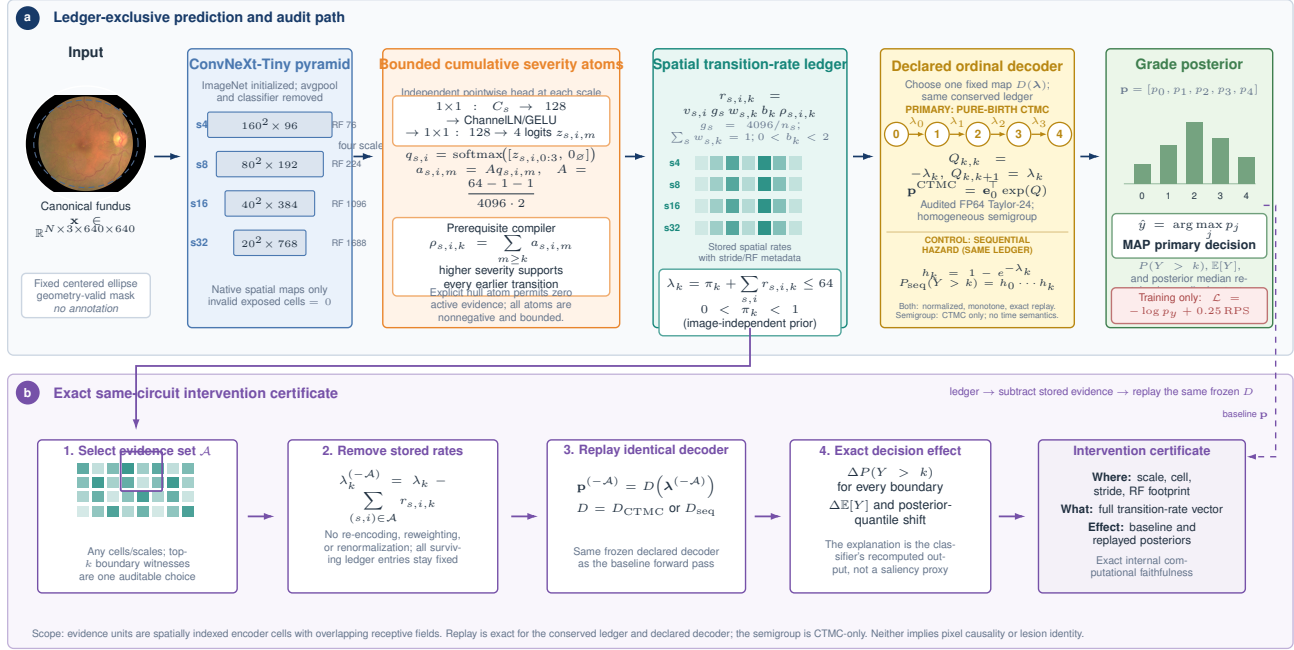


Figure 1. ORIGIN architecture. A canonical fundus image is encoded into four native ConvNeXt feature fields. Pointwise heads produce bounded severity atoms; prerequisite compilation and a conserved ledger yield four nonnegative adjacent-state rates. The primary pure-birth decoder maps those rates to the only grade posterior. For an intervention, selected stored rates are subtracted and the identical declared decoder is replayed. The certificate is exact for the model’s frozen ledger, not a claim of pixel causality or lesion identity; the replay contract also holds for the same-ledger sequential-hazard control.

3. Method

3.1. Problem and design contract

Let x_n be an image and $y_n \in \{0, \dots, K-1\}$ its ordered grade. For DR, $K = 5$ and there are $B = K - 1 = 4$ adjacent boundaries. We require that every image-dependent quantity reaching the posterior be addressable by scale and cell, nonnegative, and exactly conserved under aggregation. No global pooled classifier, auxiliary logit, or residual decision route is allowed. The full forward pass is summarized in Fig. 1.

3.2. Multiscale spatial fields

After cropping the dominant retinal field, we resize to 640×640 and apply ImageNet normalization. An ImageNet-pretrained ConvNeXt-Tiny [20], with its pooling and classifier removed, exposes four native feature fields F_s :

A fixed centered elliptical mask encodes only retinal support; it is not an annotation. A feature cell is valid when at least half of its pooled input support lies inside

Table 1. Native evidence fields for a 640×640 input. RF denotes the declared theoretical receptive-field footprint used in certificates.

Scale	Grid	Channels	Stride	RF (px)
s4	160 ²	96	4	76
s8	80 ²	192	8	224
s16	40 ²	384	16	1096
s32	20 ²	768	32	1688

the ellipse. Invalid exposed cells are zeroed after the encoder. Each scale has a separate pointwise head $1 \times 1 \rightarrow 128$, channel LayerNorm, GELU, and $1 \times 1 \rightarrow B$. There is no cross-cell operation in these heads.

3.3. Bounded atoms and prerequisite compilation

For valid cell u at scale s , the head emits logits $z_{nsu} \in \mathbb{R}^B$. Appending a fixed zero-logit null state gives

$$q_{nsum} = \frac{\exp z_{nsum}}{1 + \sum_{j=0}^{B-1} \exp z_{nsuj}}, \quad a_{nsum} = Aq_{nsum}, \quad (1)$$

for atom $m \in \{0, \dots, B-1\}$. Hence $a_{nsum} \geq 0$ and $\sum_m a_{nsum} < A$; the remaining mass is explicit “no severity evidence.”

A higher-transition atom can serve as cumulative evidence for prerequisite lower transitions. We encode this inductive bias before any image-level aggregation:

$$c_{nsuk} = \sum_{m=k}^{B-1} a_{nsum}, \quad k = 0, \dots, B-1. \quad (2)$$

Thus $c_{nsu0} \geq \dots \geq c_{nsu, B-1} \geq 0$. The atom index is a learned severity-transition type, not a named clinical lesion; semantic lesion claims require separate validation.

3.4. Conserved transition-rate ledger

Let $V_{nsu} \in \{0, 1\}$ mark valid cells and $N_{ns} = \sum_u V_{nsu}$. With reference count $R = 4096$, define

$$g_{ns} = \begin{cases} R/N_{ns}, & N_{ns} > 0, \\ 0, & N_{ns} = 0. \end{cases} \quad (3)$$

For each boundary, scale weights form a simplex, $w_{sk} = \exp \alpha_{sk} / \sum_t \exp \alpha_{tk}$, while $b_k = 2\sigma(\beta_k)$ calibrates that boundary. The stored contribution is

$$\ell_{nsuk} = V_{nsu} g_{ns} w_{sk} b_k c_{nsuk} \geq 0, \quad (4)$$

and the image rate is exactly

$$\lambda_{nk} = \pi_k + \sum_s \sum_u \ell_{nsuk}, \quad \pi_k = \sigma(\nu_k) \in (0, 1). \quad (5)$$

The prior π_k is image independent. All image-dependent influence on the posterior is contained in the spatial ledger ℓ .

Resolution-independent bound. Let the decoder ceiling be $C = 64$, the prior cap $P = 1$, the calibration cap $b_{\max} = 2$, and the roundoff reserve $\delta = 1$. We set

$$A = \frac{C - P - \delta}{Rb_{\max}} = \frac{62}{8192}. \quad (6)$$

Because geometry normalization cancels the number of valid cells and $\sum_s w_{sk} = 1$,

$$0 < \lambda_{nk} < P + Rb_{\max}A = 63 < C. \quad (7)$$

The bound holds regardless of feature-map resolution, preventing evidence from growing merely because a field contains more cells.

3.5. Pure-birth ordinal posterior

The boundary rates parameterize a $K \times K$ upper-bidiagonal generator

$$Q_{k,k} = -\lambda_k, \quad Q_{k,k+1} = \lambda_k, \quad Q_{K-1,K-1} = 0. \quad (8)$$

Starting in state zero, the unit-exposure posterior is

$$\mathbf{p} = e_0^\top \exp(Q), \quad p_j = P(Y = j \mid x). \quad (9)$$

Since Q is a valid generator, $\mathbf{p}$ is nonnegative and sums to one. We compute Eq. (9) in FP64 with a degree-24 Taylor scaling-and-squaring routine. For a decoder batch, let $\eta = \max_n \|Q_n\|_\infty$ and $s = \max\{0, \lceil \log_2(\eta/0.5) \rceil\}$. We evaluate each Taylor polynomial at $Q_n/2^s$ and square the result s times. No posterior clipping or renormalization is applied. The unit exposure is simply a fixed decoder convention; λ is latent ordinal evidence, not biological time.

The model exposes $P(Y > k) = \sum_{j=k+1}^{K-1} p_j$, expected grade $E[Y] = \sum_j j p_j$, posterior quantiles, and MAP. The frozen primary decision is $\hat{y} = \arg \max_j p_j$. Increasing all rates componentwise advances the chain in first-order stochastic order, so cumulative probabilities, expected grade, and posterior quantiles are monotone. MAP itself need not change monotonically and is not claimed to do so.

3.6. Decoder scope and sequential-hazard control

To isolate the ledger from its primary decoder, we apply adjacent conditional hazards to the identical stored rates. At horizon t , let

$$h_k(t) = 1 - \exp(-t\lambda_k), \quad P_{\text{seq}}(Y > k) = \prod_{j=0}^k h_j(t). \quad (10)$$

The corresponding normalized posterior is

$$p_y^{\text{seq}} = \begin{cases} (1 - h_y) \prod_{j < y} h_j, & y < K-1, \\ \prod_{j=0}^{K-2} h_j, & y = K-1. \end{cases} \quad (11)$$

Both decoders are normalized and coordinatewise monotone on the deployed bounded rate domain at the fixed horizon $t = 1$; both inherit exact deletion replay from the ledger. A seeded numerical inversion audit over 64 sampled interior-simplex targets found maximum posterior reconstruction errors of 2.08×10^{-12} (CTMC) and 1.95×10^{-16} (sequential). This supports coverage of those tested targets, not universal representation of every interior posterior under finite rate bounds, and gives no evidence of a fixed-horizon expressiveness advantage for the CTMC in the sampled audit.

The CTMC-specific guarantee is cross-horizon: its transition kernel $P(t) = \exp(tQ)$ shares one clock and obeys the homogeneous Markov semigroup $P(t+r) = P(t)P(r)$. The natural restartable kernel induced by Eq. (11) does not satisfy this identity in general. Our cross-sectional labels observe only one endpoint, so we treat this as a mathematical contract, not an empirically validated disease-progression claim.

Scope of guarantees. Exclusivity, nonnegative conservation, boundary addressability, and exact grouped replay are ledger-wide. Normalization and coordinatewise monotonicity hold for both declared decoder instantiations. The homogeneous shared-clock semigroup is CTMC-specific. We make no claim that the last property improves cross-sectional grading.

3.7. Exact frozen-ledger intervention

Let D_θ denote either declared deterministic decoder above. For any selected ledger set S (cells, scales, or both), deletion is

$$\lambda_{nk}^{(-S)} = \lambda_{nk} - \sum_{(s,u) \in S} \ell_{nsuk}, \quad \mathbf{p}^{(-S)} = D_\theta(\lambda_n^{(-S)}). \quad (12)$$

The complementary retain-only replay is $\lambda_{nk}^{(+S)} = \pi_k + \sum_{(s,u) \in S} \ell_{nsuk}$ and $\mathbf{p}^{(+S)} = D_\theta(\lambda_n^{(+S)})$. The encoder, support masks, geometry factors, scale weights, and all surviving ledger entries remain fixed. Therefore the reported change $\Delta P(Y > k)$ or $\Delta E[Y]$ is the literal difference between the baseline computation and the internally intervened computation. The identity follows from ledger conservation and deterministic replay; it is not unique to $\exp(Q)$. This avoids a common confound of pixel occlusion: replacing image content and re-encoding can create out-of-distribution artifacts and can change every feature.

Each certificate stores the scale, cell coordinate, receptive-field footprint, full boundary-rate vector, baseline posterior, replayed posterior, and decision effect. Overlapping receptive fields mean that cells are computational units, not disjoint pixel segments. Accordingly, exactness establishes computational faithfulness inside Origin; it does not by itself prove pixel causality, lesion identity, clinical sufficiency, or clinical necessity. For budget-matched audits we define the nominal stride-area cost $A_{\text{nom}}(S) = \sum_{(s,u) \in S} \text{stride}_s^2$. This is an additive multiscale addressability cost, not unique pixel coverage or the union area of overlapping receptive fields.

3.8. Objective

We train the generated posterior with negative log likelihood and the ranked probability score (RPS), a proper score sensitive to ordinal distance [10, 11]:

$$\mathcal{L} = -\frac{1}{N} \sum_n \log p_{n,y_n} + \frac{0.25}{N(K-1)} \sum_{n,k} (P(Y_n > k) - \mathbf{1}[y_n > k])^2. \quad (13)$$

Here $\mathbf{1}$ is implemented as a binary target; no separate head is used. There is no class weighting, oversampling, auxiliary classifier, or active evidence-budget penalty in the frozen model.

4. Experiments

4.1. Datasets and locked protocol

EyePACS. The Kaggle 2015 DR training set contains 35,126 images with grade counts [25,810, 2,443, 5,292, 873, 708] [9]. We use 10-fold outer cross-validation grouped by patient identifier parsed from the filename; left and right eyes from one patient remain together in outer and inner partitions.

APTOS 2019. APTOS contains 3,662 images with grade counts [1,805, 370, 999, 193, 295], acquired under heterogeneous cameras and imaging conditions [2]. Because patient identifiers are unavailable, we use image-stratified 5-fold outer cross-validation and state this weaker split explicitly.

Within each outer training pool, 10% is reserved as an inner validation split. Training and checkpoint choice use only the inner split. After selection is finalized, the checkpoint is evaluated on the locked outer fold; a second deterministic pass exports per-image probabilities for auditing. Neither pass affects training or selection. Architecture and hyperparameters were developed on fixed inner-validation splits. The bounded V3 architecture was frozen on 10 September 2026; the immutable full-CV launcher, split policy, and selection rule were frozen on 22 September before the 15 outer-fold evaluations. Thus per-fold checkpointing is nested, but architecture selection was not independently repeated inside every outer fold. No outer test fold was used for checkpoint selection. The final EyePACS OOF audit contains 35,126 unique images from 17,563 patients (two eyes each), with no patient crossing an outer fold.

4.2. Implementation and metrics

We optimize with AdamW (encoder LR 10^{-4} , new-head LR 5×10^{-4} , weight decay 10^{-5}), batch size 8,

and gradient clipping at 5. The encoder is frozen for two epochs and then jointly fine-tuned. ReduceLROnPlateau monitors validation loss (factor 0.2, patience 5, minimum LR 10^{-6}). APTOS runs for at most 35 epochs with early-stop patience 12; EyePACS uses 75 and 15. The checkpoint rule is lexicographic: validation accuracy, then QWK, then lower loss. Training uses horizontal/vertical flips, rotations up to 30° , and moderate color jitter. AMP is used for the encoder and heads; the generator is FP64. Runs used Python 3.11, PyTorch 2.5.1, and a 32-GB Tesla V100.

We report accuracy, discrete MAP-grade MAE, posterior-expected-grade MAE, QWK, balanced accuracy, macro-F1, 15-bin top-label expected calibration error (ECE), confusion matrices, and per-grade recall. Fold spread is the sample standard deviation. Class MAP is the primary decision rule fixed before looking at outer tests; alternative decoders are not selected post hoc.

Numerical reference audit. The deployed FP64 Taylor scaling-and-squaring decoder was compared with an independent 90-decimal-digit mpmath.expm reference on 138 seeded random and adversarial rate vectors in $[0, 64]^4$, plus 12 independent finite-difference gradient cases. The audit passed without clipping or renormalization: maximum absolute posterior, row-sum, and absolute-gradient errors were respectively below 1.5×10^{-14} , 1.7×10^{-14} , and 1.4×10^{-15} .

Artifacts and verification. An automated release manifest binds all 15 selected checkpoints and 15 compressed split-membership tables to SHA-256 hashes, split signatures, implementation signatures, and the source commit. The anonymous release gate also requires OOF posteriors, deterministic split construction, intervention records, metric/bootstrap scripts, and restart provenance to pass checksums before any newly added acceptance-audit table is reported. Membership identities are hashed and no fundus pixels are redistributed; image access remains subject to the source dataset licenses.

4.3. Required comparisons

The external numbers in Tab. 2 are literature-reported rather than matched reimplementations. Exact fold membership and checkpoint-selection artifacts are unavailable for CORE and Ord2Seq, while the Hybrid study does not publish fold assignments or fold-wise dispersion. To avoid mixing estimators, we report both our primary discrete MAP-grade MAE and a posterior-expected-grade MAE matched to Ord2Seq’s definition. Protocol and preprocessing differences can still move

results substantially; therefore the external rows contextualize, but do not establish, superiority. Studies using materially different fold counts or split construction are omitted rather than juxtaposed numerically. Our matched fold-9 study trains pooled multiscale softmax, cumulative-logit, and conditional continuation-ratio heads; a same-ledger sequential-hazard decoder; direct-simplex and independent-sigmoid atom variants; NLL-only, fine-only, and coarse-only variants; and a contemporaneously rerun full Origin model. These pooled ordinal forms are matched controls, not exact reimplementations of published CORAL or CORN systems. Matched CLOC and sparse/additive local models were not part of that retrospective suite. A separately frozen acceptance matrix now reserves comparisons against the same-ledger hazard, pooled continuation ratio, and matched in-repo analogues of ordinal Additive MIL and a nonnegative sparse local-logit model across matched folds and seeds. These are controlled architectural analogues, not claims of reproducing the original authors’ training code. Their results are withheld until the complete release gate passes.

4.4. Retrospective matched fold-9 mechanism study

We separately study mechanisms on EyePACS outer fold 9. The final V3 architecture predates full CV, but the mechanism protocol was added on 26 September after the full-CV aggregate and fold 9 had been observed. This is therefore retrospective evidence, not a confirmatory test. The split contains 28,456 training, 3,162 inner-validation, and 3,508 locked outer-test images. The test images form 1,754 patient clusters (both eyes remain paired). Ten independently trained variants use the same patient split, ImageNet initialization policy, augmentation, optimizer, dedicated training-stream seed, checkpoint rule, and maximum 75-epoch training policy. Inner validation alone selects each checkpoint. After all ten training-completion markers existed, one nonadaptive suite-level release evaluated every selected checkpoint on the locked outer fold. The study uses one training seed, so its patient-cluster intervals quantify test-cohort sampling variation conditional on fitted models, not training variability.

5. Results

5.1. Grading performance

On complete APTOS outer CV, Origin obtains $84.872 \pm 0.508\%$ fold-mean accuracy, 0.9117 ± 0.0073 QWK, and 0.1936 ± 0.0072 MAP-grade MAE. Pooled out-of-fold accuracy is 84.872%, with 63.67% balanced accuracy, 0.6438 macro-F1, and 0.0502 ECE. The pooled posterior-expected-grade MAE is 0.2168 (fold mean

Table 2. Outer-cross-validation results. ORIGIN values are pooled out-of-fold predictions from our locked protocol; external values are quoted from the cited primary papers and are not matched reruns. Reported protocols are not split-identical. ^M denotes discrete MAP-grade MAE and ^E posterior-expected-grade MAE; unmarked external values retain the estimator reported by their authors. Higher accuracy/QWK and lower MAE are better.

Dataset	Protocol	Method	Acc. (%)	QWK	MAE (est.)	Status	Source
EyePACS	reported 10-fold	CORE+POE (VGG-19)	84.3	—	0.2718	reported	[17]
	reported 10-fold	Ord2Seq (PVTv2-B1)	84.2±0.5	—	0.25±0.07 ^E	fold mean	[29]
	reported 10-fold	Hybrid Eff.-V2S	83.8	—	0.21	reported	[23]
	patient 10-fold	ORIGIN	85.99	0.8153	0.1963 ^M /0.2496 ^E	pooled OOF	ours
APTOS	image 5-fold	ORIGIN	84.87	0.9118	0.1936 ^M /0.2168 ^E	pooled OOF	ours

Matched controls on one previously observed EyePACS fold are summarized in Sec. 6 and reported fully in the supplement; they do not replace the complete-CV estimate or establish population-level superiority.

Table 3. Per-grade recall (%) from complete pooled outer-fold predictions.

Dataset	G0	G1	G2	G3	G4
EyePACS (10/10)	96.91	28.53	71.69	35.28	55.37
APTOS (5/5)	98.78	50.27	91.89	7.25	70.17

0.2168±0.0080). The small accuracy spread indicates stable overall grading across the five partitions.

On complete EyePACS outer CV, pooled out-of-fold performance is 85.988% accuracy, 0.8153 QWK, 0.1963 MAP-grade MAE, 57.56% balanced accuracy, 0.6223 macro-F1, and 0.0288 ECE. Fold-mean accuracy is 85.988±0.442%; fold-mean QWK and MAP-grade MAE are 0.8152±0.0189 and 0.1963±0.0113. The estimator-matched posterior-expected-grade MAE is 0.2496 (fold mean 0.2496±0.0143), essentially the same as Ord2Seq’s reported 0.25±0.07. Among the audited studies labelled as 10-fold EyePACS evaluations, our accuracy is descriptively 1.69 points above CORE+POE (84.3%) and 1.79 points above Ord2Seq-PVT (84.2%). These are unpaired, protocol-heterogeneous differences, not matched evidence of superiority.

Long-tail behavior. Table 3 prevents headline accuracy from hiding failure modes. The no-DR class is recognized reliably, but mild and severe NPDR remain hard. On EyePACS, 1,335 of 2,443 grade-1 images are predicted as grade 0 and 500 of 873 grade-3 images as grade 2. Overall, 94.96% of predictions are exact or adjacent, while 5.04% differ by at least two grades. APTOS grade-3 recall is only 7.25%: 136 of 193 grade-3 images are predicted as grade 2. These are major current limitations, not evidence that the ordinal mechanism has solved imbalance. Importantly, the frozen objective intentionally uses the unweighted population

likelihood; targeted imbalance methods must be compared without compromising calibration or the ledger semantics.

5.2. Structural identity versus empirical utility

The intervention identity in Eq. (12) is structural. The ten EyePACS folds archive 40 inner-validation certificates (four per fold): the maximum stored-ledger partition error is exactly zero and the maximum total-rate replay error is 5.56×10^{-17} . The present sample is not grade-stratified—34 of 40 cases are grade 0—and no single-cell deletion changes MAP. Moreover, 29 of 40 selected witnesses come from the coarse s16 or s32 fields, and mean $|\Delta E[Y]|$ is only 0.00389. These artifacts verify replay arithmetic; they do not demonstrate useful fine-grained localization. We therefore separate the proved arithmetic identity of frozen-ledger replay from the still-open empirical question of whether learned witnesses are local, useful, or semantically aligned.

IDRiD mask association. All ten independently inner-selected EyePACS outer-fold checkpoints evaluate the 81-image IDRiD segmentation subset [22], with no IDRiD training, tuning, or selection. We form the official MA/HE/EX/SE union; apply the image-derived dominant-FOV crop to image and masks; and resize the image bilinearly and mask by nearest neighbor. A native-lattice bin is positive iff it contains any mask pixel; its score is the stored local_rate_map at true-grade boundary $b(y) = \min\{\max(y-1, 0), 3\}$. AP is computed per image×checkpoint×scale, then averaged over checkpoints/scales within image; inference clusters by image. Deletion uses 20 seeded, deterministic non-lesion controls per image, checkpoint, and scale.

V1 fine deletion was exactly count-matched and yielded mask-guided replay lift 0.0111 in $|\Delta E[Y]|$ ([0.0097, 0.0125]). Its AP and coarse/all-scale deletion estimates are provisional: AP did not group

532	exact score ties and 14 coarse image×scale cases	retrospective fold, one seed, and multiple unadjusted	579
533	were under-matched. A sealed corrective v2 (pending)	comparisons, these are mechanism findings only.	580
534	uses threshold-grouped AP and 20 paired within-		
535	image/scale draws of equal unique lesion/nonlesion	7. Discussion and Limitations	581
536	counts, matching scale composition, geometry exposure,	What is novel. The ingredients—ConvNeXt features,	582
537	and the pre-atom boundary multiplier. Retinal	nonnegative maps, ordinal decoders, generators, and	583
538	anatomy and removed rate mass are not matched; the	proper scores—are established. The contribution is	584
539	latter is the outcome. These are internal ledger interventions,	their coupling into a boundary-indexed ledger that is	585
540	not pixel deletion, lesion recognition, or	the exclusive image path, cumulative across prerequisites,	586
541	clinical necessity.	and exactly group-replayable. This narrow, falsifiable	587
		claim holds for both declared decoders.	588
542	6. Ablations and Falsification Tests		
543	Ablations should be able to falsify, rather than merely	What exactness does not imply. Deletion asks what	589
544	decorate, an architectural claim. We therefore compare	the model would predict without selected stored entries	590
545	the contemporaneously rerun full model against	while survivors remain fixed. It does not establish pixel	591
546	nine controls under the frozen protocol in Sec. 4.4.	causality, lesion identity, or clinical usefulness. Coarse	592
547	All per-image posteriors, absolute metrics, and 10,000	receptive fields make this distinction especially important;	593
548	patient-cluster paired intervals were reproduced independently.	pixel masking, external masks, and clinician evaluation	594
549	The complete ten-arm tables are placed in	remain complementary.	595
550	the supplement because this retrospective single-fold		
551	study is mechanism evidence rather than the primary	Data and protocol. EyePACS is strongly imbalanced,	596
552	benchmark.	APTOS lacks patient identifiers, and both use single	597
		noisy competition labels. Architecture search used a	598
553	Predictive controls. The pooled continuation-ratio	fixed inner split rather than fully nested hyperparameter	599
554	model is the strongest grader on this fold: 86.40%	selection, so CV estimates performance of the frozen	600
555	accuracy, 0.8299 QWK, and 0.1861 MAP-MAE, versus	configuration, not clinical deployment. External domains	601
556	85.66%, 0.8132, and 0.2007 for full Origin. Accuracy	and multi-reader labels are needed. Two non-	602
557	and QWK intervals include zero, but its MAP-MAE	finite attempts failed closed and were rerun unchanged;	603
558	is lower by 0.0145 ($[-0.0274, -0.0017]$); conversely,	restart provenance is supplemental. The revision manifest	604
559	ECE is worse by 0.0273 ($[0.0198, 0.0357]$). This	hashes checkpoints and split memberships for integrity	605
560	performance–calibration–auditability tradeoff does not	and disjointness checks; hashes alone provide	606
561	establish population-level superiority from one fold.	neither the artifacts nor executable re-inference.	607
562	Scale falsification. Fine fields alone are decisively	Model scope. The ImageNet-pretrained model is not	608
563	insufficient for grading: relative to full Origin they lose	a medical foundation model. Evidence atoms are	609
564	1.97 accuracy points ($[-2.88, -1.08]$) and 0.0589 QWK	severity-indexed, not clinical concepts; generalization	610
565	($[-0.084, -0.035]$); grade-4 recall falls from 66.15%	beyond fundus grading requires new modalities and	611
566	to 21.54%. Coarse-only evidence has the same accuracy	datasets.	612
567	point estimate as full Origin, with all primary intervals		
568	spanning zero. This is not equivalence, but it prevents	Potential impact. Replay can audit shortcuts only if	613
569	a claim of demonstrated fine-lesion localization.	the pending controlled benchmark passes; it does not	614
		replace clinical judgment. Rare-grade errors require	615
570	Decoder, atoms, and objective. The same-ledger	external validation, calibration, monitoring, and human	616
571	sequential-hazard decoder, direct-simplex atoms,	oversight.	617
572	independent-sigmoid atoms, and NLL-only objective	8. Conclusion	618
573	all have accuracy, QWK, and MAP-MAE intervals	Origin makes a conserved boundary-indexed ledger the	619
574	spanning zero relative to full Origin. Thus this fold	exclusive prediction input and an exactly replayable	620
575	does not show that the CTMC, prerequisite compiler,	audit object. Complete CV supports competitive grading;	621
576	or RPS improves grading. Sequential hazards improve	a retrospective fold shows neither an advantage over	622
577	ECE by 0.0072 ($[0.0003, 0.0137]$). All variants contain	pooled heads nor fine-only sufficiency. Broader claims	623
578	27.86–28.07M parameters. Because this is one	await the sealed audits.	624

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